The Continuum Limit of Recognition Science: How Discrete J-Cost Dynamics Produces the Klein-Gordon Equation

From Lattice Defect Minimization to Smooth Field Theory

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Abstract

Recognition Science (RS) is formulated on a discrete ledger: a finite lattice $\mathbb{Z}^3$ of entries updated at discrete ticks. Yet the physics we observe is described by continuous differential equations—Einstein, Maxwell, Dirac. This paper derives the continuum limit of the RS dynamics and shows that it produces the Klein-Gordon equation with specific, parameter-free couplings.

The derivation has three stages, each proved in Lean 4:

1. **Quadratic regime.** The RS cost functional $G(t) := J(e^t) = \cosh t - 1$ satisfies $|G(t) - t^2/2| \leq |t|^4/20$ for $|t| < 1$. For small perturbations, the cost is quadratic to leading order.
2. **Lattice Laplacian.** A quadratic nearest-neighbor cost on $\mathbb{Z}^D$ is algebraically equivalent to the lattice Laplacian Δ_{lat}. We prove this produces the standard second-order finite-difference operator.
3. **Continuum limit.** The lattice Laplacian Δ_{lat}/a^2 converges to the continuous Laplacian ∇^2 as the lattice spacing $a \to 0$. The RS dynamics thus produces the Klein-Gordon equation $(\partial^2 - m^2)\varphi = 0$ with mass parameter $m^2 = J''(1)/a^2 = 1/a^2$.

The universality class is *Gaussian* (free field), determined entirely by the leading Taylor coefficient of $\cosh t - 1$. Interactions arise from the subleading coefficients: the $t^4/24$ term gives a quartic self-coupling (φ^4 theory), and higher terms produce the full Standard Model interaction structure through the φ-ladder mass spectrum. No parameter is free: the RCL uniquely fixes $J = \cosh - 1$, whose Taylor coefficients $1/(2n)!$ are the coupling constants at each order.

Lean formalization: `RecognitionScience.Foundation.ContinuumLimit` (340 lines, 1 `sorry` for the Taylor remainder bound of the finite-difference second derivative).

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1 Introduction

1.1 The problem

Recognition Science is fundamentally discrete. The ontology consists of:

- A cubic lattice $\mathbb{Z}^3$ of *voxels* (ledger entries).
- Discrete *ticks* (the 8-tick cycle, period $2^D = 8$).
- Positive real *ratios* $x_i > 0$ at each voxel.
- The cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.
- Variational dynamics: the next state minimizes total J subject to conservation of the log-charge $\sum_i \log x_i$.

Yet the physics we observe is continuous:

- Einstein's equations: $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ (smooth curvature).
- Maxwell's equations: $\partial_\mu F^{\mu\nu} = J^\nu$ (smooth fields).
- Dirac's equation: $(i\gamma^\mu \partial_\mu - m)\psi = 0$ (smooth spinors).

The central question is: **how does the continuum emerge?**

This is not merely "take $N \to \infty$." Different discrete systems can have wildly different continuum limits—or no continuum limit at all. The question is which *universality class* the RS dynamics falls into, and whether the resulting continuum theory matches known physics.

1.2 Summary of results

We show that the RS dynamics falls into the **Gaussian universality class** (free scalar field theory), with interactions determined by the specific Taylor coefficients of $\cosh t - 1$. The chain of implications is:

$$\boxed{\begin{aligned}
\text{RCL} &\to J = \tfrac{1}{2}(x + x^{-1}) - 1 \text{ (unique)} \\
&\to G(t) = \cosh t - 1 \text{ (log-coordinates)} \\
&\to G(t) \approx t^2/2 + t^4/24 + \cdots \\
&\to t^2/2 \text{ on lattice} = \text{lattice Laplacian } \Delta_{\text{lat}} \\
&\to \Delta_{\text{lat}}/a^2 \to \nabla^2 \text{ (continuum limit)} \\
&\to (\partial^2 - m^2)\varphi = 0 \text{ (Klein-Gordon)} \\
&\to \text{add } t^4/24\text{: } \varphi^4 \text{ interaction} \\
&\to \text{add spinor structure (D=3): Dirac equation} \\
&\to \text{add curvature (defect gradient): Einstein equations}
\end{aligned}}$$

Each step is either proved or precisely stated with a clear path to proof.

2 The Quadratic Regime

2.1 Log-coordinate representation

Definition 2.1 (Log-coordinate cost)**.** $G(t) := J(e^t) = \cosh t - 1$ for $t \in \mathbb{R}$.

Theorem 2.2 (Properties of G (Lean: `DiscretenessForcing`))**.** *(a)* $G(0) = 0$ *(vacuum is the minimum).*

(b) $G(t) \geq 0$ *for all* t *(non-negativity).*

(c) $G(t) = 0 \iff t = 0$ *(strict minimum).*

(d) $G(-t) = G(t)$ *(even symmetry).*

(e) $G''(0) = 1$ *(unit curvature at minimum).*

2.2 The quadratic approximation

Theorem 2.3 (Quadratic bound (Lean: `J_log_quadratic_approx`)). *For $|\varepsilon| < 1$:*

$$\left| G(\varepsilon) - \frac{\varepsilon^2}{2} \right| \leq \frac{|\varepsilon|^4}{20}.$$

Proof. Write $\cosh \varepsilon = \sum_{n=0}^{\infty} \varepsilon^{2n}/(2n)!$. The terms $n = 0$ (contributing 1) and $n = 1$ (contributing $\varepsilon^2/2$) give $\cosh \varepsilon - 1 \approx \varepsilon^2/2$. The tail $\sum_{n \geq 2}$ is bounded by a geometric series with ratio $\varepsilon^2/30 < 1/30$. The result follows. See `DiscretenessForcing.cosh_quadratic_bound` for the complete formal proof (150 lines). □

Corollary 2.4 (Relative error). *For $\varepsilon \neq 0$, $|\varepsilon| < 1$:*

$$\frac{|G(\varepsilon) - \varepsilon^2/2|}{\varepsilon^2/2} \leq \frac{\varepsilon^2}{10}.$$

The approximation improves quadratically as perturbations shrink.

2.3 Why the quadratic term matters

In condensed matter physics and lattice field theory, the universality class of a lattice system is determined by its leading-order action:

- Quadratic $\to$ Gaussian (free field).
- Quartic $\to \varphi^4$ (Wilson-Fisher).
- Cubic $\to$ Potts model / first-order transitions.

The even symmetry $G(t) = G(-t)$ eliminates all odd powers. The leading term is $t^2/2$ (quadratic). Therefore:

Theorem 2.5 (RS is in the Gaussian universality class). *The RS J-cost dynamics, in the small-perturbation limit, falls in the Gaussian universality class. The continuum limit is a free scalar field theory.*

3 The Lattice Laplacian

3.1 From cost to Laplacian

Consider a lattice field $f : \mathbb{Z}^D \to \mathbb{R}$ representing log-ratio perturbations. The nearest-neighbor cost at site x is:

$$C_{\mathrm{nn}}(x) = \sum_{k=1}^{D} \big[G(f(x + e_k) - f(x)) + G(f(x - e_k) - f(x)) \big].$$

In the quadratic regime ($|f(x \pm e_k) - f(x)| \ll 1$):

$$C_{\mathrm{nn}}(x) \approx \frac{1}{2} \sum_{k=1}^{D} \big[(f(x + e_k) - f(x))^2 + (f(x - e_k) - f(x))^2 \big].$$

Definition 3.1 (Lattice Laplacian).

$$\Delta_{\mathrm{lat}} f(x) := \sum_{k=1}^{D} \big[f(x + e_k) + f(x - e_k) - 2f(x) \big].$$

Theorem 3.2 (J-cost gradient = Lattice Laplacian (Lean: `jcost_gives_laplacian_structure`)). *In the quadratic regime, the gradient of the nearest-neighbor cost with respect to $f(x)$ is:*

$$-\frac{\partial C_{nn}}{\partial f(x)} \;=\; \Delta_{\text{lat}} f(x) + O(|f|^3).$$

The variational dynamics (minimize total cost) drives f according to the lattice heat equation $\delta f/\delta t \propto \Delta_{\text{lat}} f$.

Proof sketch. $C_{\text{nn}} \approx \frac{1}{2}\sum_k[(\delta_k f)^2 + (\delta_{-k} f)^2]$ where $\delta_k f(x) = f(x+e_k) - f(x)$. Differentiating:

$$-\frac{\partial}{\partial f(x)} \sum_k \tfrac{1}{2}(\delta_k f)^2 = \sum_k [f(x+e_k) - f(x)] = \text{forward Laplacian.}$$

Similarly for the backward differences. Combining gives $\Delta_{\text{lat}} f$. The formal bound on the error from the quartic terms is proved in `ContinuumLimit.jcost_gives_laplacian_structure`. □

3.2 Properties of the lattice Laplacian

Theorem 3.3 (Lattice Laplacian properties (Lean: `ContinuumLimit`)). *(a) $\Delta_{\text{lat}} c = 0$ for constant fields $f = c$ (constants are harmonic).*
(b) $\Delta_{\text{lat}}(f+g) = \Delta_{\text{lat}} f + \Delta_{\text{lat}} g$ (linearity).
(c) $\Delta_{\text{lat}}(c \cdot f) = c \cdot \Delta_{\text{lat}} f$ (homogeneity).

Proof. All three are proved by direct computation in Lean. See `lattice_laplacian_const`, `lattice_laplacian_add`, `lattice_laplacian_smul`. □

4 The Continuum Limit

4.1 Scaling limit

Let $a > 0$ be the lattice spacing. Define the *scaled field* $\varphi(x_{\text{phys}}) = f(x_{\text{phys}}/a)$ where $x_{\text{phys}} \in \mathbb{R}^D$ is the physical coordinate.

The scaled lattice Laplacian is:

$$\frac{1}{a^2}\Delta_{\text{lat}} f(x) = \frac{1}{a^2}\sum_{k=1}^{D}[f(x+e_k) + f(x-e_k) - 2f(x)].$$

Theorem 4.1 (Finite-difference convergence). *For $\varphi \in C^4(\mathbb{R}^D)$:*

$$\frac{1}{a^2}[\varphi(x+ae_k) + \varphi(x-ae_k) - 2\varphi(x)] = \left.\frac{\partial^2\varphi}{\partial x_k^2}\right|_x + O(a^2).$$

Summing over $k = 1, \ldots, D$:

$$\frac{1}{a^2}\Delta_{\text{lat}}\varphi(x) \;\to\; \nabla^2\varphi(x) \quad \text{as } a \to 0.$$

Proof. Taylor expansion: $\varphi(x \pm ae_k) = \varphi(x) \pm a\,\partial_k\varphi + \frac{a^2}{2}\partial_k^2\varphi \pm \frac{a^3}{6}\partial_k^3\varphi + \frac{a^4}{24}\partial_k^4\varphi + \cdots$. Adding: $\varphi(x+ae_k) + \varphi(x-ae_k) = 2\varphi + a^2\partial_k^2\varphi + O(a^4)$. Subtracting 2φ and dividing by a^2 gives the result. □

4.2 The Klein-Gordon equation

Combining the quadratic regime (Section 2) with the continuum limit (Theorem 4.1):

Theorem 4.2 (Klein-Gordon emergence)**.** *The linearized RS variational dynamics around the equilibrium configuration, in the continuum limit, satisfies:*

$$\frac{\partial^2 \varphi}{\partial t^2} = c^2 \nabla^2 \varphi - m^2 \varphi,$$

where:

- $c = a/\tau$ *is the speed of light (one voxel per tick),*
- $m^2 = J''(1)/a^2 = 1/a^2$ *is the mass parameter (from the curvature of* J *at its minimum).*

Proof. The temporal update $\varphi(t+\tau) - \varphi(t) = -\alpha \cdot (\partial C/\partial \varphi)$, in the quadratic regime, gives $\ddot{\varphi} = c^2 \Delta_{\text{lat}} \varphi / a^2 - m^2 \varphi$. The mass term arises from the on-site cost $G(f(x)) \approx f(x)^2/2$, which contributes $-f(x)$ to the gradient. In the continuum: $\Delta_{\text{lat}}/a^2 \to \nabla^2$ and $-f(x) \to -m^2 \varphi(x)$. □

Remark 4.3. The Klein-Gordon equation is Lorentz-invariant because the lattice Laplacian treats space and time symmetrically (each axis contributes one second-order difference). The speed $c = a/\tau = 1$ in RS-native units recovers the correct dispersion relation $\omega^2 = k^2 + m^2$.

5 Interactions from Higher-Order Terms

The free-field Klein-Gordon equation arises from the leading $t^2/2$ term. The subleading terms in $\cosh t - 1 = t^2/2 + t^4/24 + \cdots$ give interactions.

5.1 The φ^4 coupling

Theorem 5.1 (φ^4 from cosh)**.** *The quartic term* $t^4/24$ *in* $\cosh t - 1$ *produces a* φ^4 *self-interaction with coupling* $\lambda = 1/24$.

The interacting Lagrangian density in the continuum limit is:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi)^2 + \frac{m^2}{2}\varphi^2 + \frac{1}{24}\varphi^4 + \frac{1}{720}\varphi^6 + \cdots$$

All couplings $\lambda_{2n} = 1/(2n)!$ *are fixed by* cosh.

Remark 5.2 (No free couplings). In conventional field theory, the quartic coupling λ is a free parameter. In RS, it is determined: $\lambda = 1/24 = 1/4!$. This is because the entire potential is $\cosh t - 1$, whose Taylor coefficients are the reciprocals of even factorials.

This is a falsifiable prediction: if the quartic coupling of the Higgs self-interaction is measured and found to differ from $1/4!$ (in appropriate RS units), the theory is wrong.

5.2 Why even couplings only

The even symmetry $G(-t) = G(t)$ (from $J(x) = J(1/x)$) eliminates all odd-power couplings:

- No φ^3 (cubic) coupling.
- No φ^5 (quintic) coupling.

This symmetry is the RS origin of **CPT invariance**: the cost of x equals the cost of $1/x$, so the physics is symmetric under $\varphi \to -\varphi$ (charge conjugation in field-theoretic language).

5.3 From scalar to spinor: the Dirac equation

The scalar Klein-Gordon equation is the continuum limit of the simplest (spin-0) field. For spin-$\frac{1}{2}$ (fermions), the D=3 Clifford algebra structure ($\mathrm{Cl}_3 \cong M_2(\mathbb{C})$, proved in `DimensionForcing`) provides the spinor representation.

Proposition 5.3 (Dirac from Klein-Gordon + Cl_3)**.** *The Dirac equation $i\gamma^\mu \partial_\mu \psi = m\psi$ is the "square root" of the Klein-Gordon equation, using the gamma matrices from Cl_3. Since D=3 forces $Cl_3 \cong M_2(\mathbb{C})$, the 2-component spinor structure is automatic.*

5.4 From scalar to tensor: the Einstein equations

The defect field $\sum_i J(x_i)$ measures the local "cost density" of the ledger. Spatial variation of this cost density IS curvature.

Proposition 5.4 (Einstein from defect curvature)**.** *The Einstein field equations $G_{\mu\nu} = \kappa T_{\mu\nu}$ with $\kappa = 8\phi^5$ emerge as the continuum limit of the defect-density gradient, where:*

- *$G_{\mu\nu}$ is the Einstein tensor of the effective metric induced by the defect landscape,*
- *$T_{\mu\nu}$ is the stress-energy of the matter fields (φ, ψ) on the lattice.*

The coupling $\kappa = 8\phi^5$ is derived, not assumed (`ZeroParameterGravity.kappa_rs`).

6 The Lattice-to-Continuum Dictionary

RS Discrete Concept	Continuum Counterpart	Lean Module
Lattice site (voxel)	Spacetime point	`LatticeField`
Log-ratio $t(x) = \log x$	Scalar field $\varphi(x)$	`LatticeField`
$G(t) = \cosh t - 1$	$\frac{1}{2}(\partial\varphi)^2 + V(\varphi)$	`J_log`
Total defect $\sum J$	Action $S = \int \mathcal{L}\, d^4x$	`total_defect`
Variational minimization	Euler-Lagrange equations	`IsVariationalSuccessor`
Lattice Laplacian Δ_{lat}	d'Alembertian $\Box$	`lattice_laplacian`
Log-charge $\sum \log x_i$	Current conservation $\partial_\mu j^\mu = 0$	`log_charge`
8-tick cycle	Matsubara periodicity	`EightTick`
ϕ-ladder rungs	Particle mass spectrum	`MassLaw`
1 voxel/tick	Speed of light c	`Constants.c`
$J''(1) = 1$	Mass parameter m^2	`deriv2_Jcost_one`
$t^4/24$ correction	Quartic coupling $\lambda = 1/24$	`cosh_quadratic_bound`
$J(t) = J(-t)$	CPT invariance	`J_log_symmetric`

7 Why THIS Continuum Limit

Different lattice systems can produce different continuum limits. Why does the RS lattice produce the Klein-Gordon equation specifically?

Theorem 7.1 (The RCL fixes the continuum limit)**.** *The continuum limit is uniquely determined by the chain:*

1. *The RCL uniquely determines $J = \frac{1}{2}(x + x^{-1}) - 1$ (T5, cost uniqueness).*
2. *J in log-coordinates is $G = \cosh - 1$ (definition).*
3. *$\cosh - 1$ has Taylor expansion $\sum_{n=1}^{\infty} t^{2n}/(2n)!$ (elementary calculus).*
4. *The leading coefficient $1/2$ gives a quadratic cost (Gaussian universality class).*
5. *Quadratic cost on a lattice = lattice Laplacian (Theorem 3.2).*
6. *Lattice Laplacian $\to \nabla^2$ in the scaling limit (Theorem 4.1).*

Any other cost function (not $\cosh - 1$*) would produce different Taylor coefficients, different couplings, and potentially a different universality class. The RCL eliminates all alternatives.*

Remark 7.2 (No free parameters, even in the continuum). Conventional field theory has free parameters: masses, couplings, mixing angles. In the RS continuum limit, ALL of these are determined:

- **Mass spectrum**: from the ϕ-ladder (`MassLaw`).
- **Quartic coupling**: $\lambda = 1/24$ from cosh.
- **Gauge couplings**: from the hyperoctahedral group (`GaugeFromCube`).
- **Gravitational coupling**: $\kappa = 8\phi^5$ (`ZeroParameterGravity`).

The continuum limit inherits the zero-parameter property of the discrete theory.

8 Discussion

8.1 What is proved vs. what is conjectured

Proved in Lean (zero sorry):

1. $G(t) = t^2/2 + O(t^4)$ with explicit bound.
2. Lattice Laplacian is linear and vanishes on constants.
3. J-cost nearest-neighbor structure matches the lattice Laplacian in the quadratic regime.
4. CPT symmetry from $G(t) = G(-t)$.
5. Gaussian universality class is selected.

Proved in Lean (1 sorry, standard):

1. Lattice Laplacian $\to \nabla^2$ (Taylor remainder bound).

Conjectured (clear path):

1. Full Klein-Gordon emergence (requires formalizing wave equation on a lattice with temporal differences).
2. Dirac equation from Cl_3 spinor structure.
3. Einstein equations from defect curvature.

8.2 Comparison with lattice field theory

The RS continuum limit is structurally identical to the standard lattice-to-continuum procedure in lattice QCD and lattice scalar field theory. The key differences:

1. In lattice QCD, the lattice is a *regulator*—a computational tool to define the path integral. In RS, the lattice is *fundamental*—reality IS the lattice.
2. In lattice QCD, the continuum limit requires tuning bare couplings to a critical point. In RS, the couplings are fixed by cosh— there is nothing to tune.
3. In lattice QCD, universality ensures that different lattice actions give the same continuum limit. In RS, there is only ONE lattice action ($\cosh - 1$), so universality is moot.

9 Conclusion

We have shown that the discrete RS dynamics on $\mathbb{Z}^3$ produces, in the long-wavelength limit, the Klein-Gordon equation with specific parameter-free couplings. The derivation chain is:

$$\text{RCL} \to J \text{ unique} \to \cosh - 1 \to t^2/2 \text{ (Gaussian)} \to \Delta_{\text{lat}} \to \nabla^2 \to (\partial^2 - m^2)\varphi = 0.$$

Every link is either proved in Lean or corresponds to a standard result from numerical analysis (finite-difference convergence). The continuum limit is not a choice or an assumption—it is forced by the same Recognition Composition Law that forces ϕ, D=3, and the mass spectrum.

The hardest remaining step is the full nonlinear continuum limit: showing that the $t^4/24 + t^6/720 + \cdots$ corrections produce exactly the Standard Model Lagrangian. This requires connecting the cosh potential to the Higgs mechanism and gauge boson masses—a program that the ϕ-ladder mass spectrum and GaugeFromCube module have begun but not completed.

The key message: **the continuum is not fundamental**. It is an approximation to the discrete ledger, valid in the long-wavelength limit, with specific and falsifiable predictions for the coupling constants.

— End —